An Adaptive Local Joint-Weighted Method for Continuous Stripping Intensity Correction of Airborne Bathymetric LiDAR

Xue Ji, Zhen Dong, *Member, IEEE*, Zheng Wei, Mingchang Wang, Wenxue Xu, and Lin Zhang, *Member, IEEE*

***Abstract*— Alongside geometric information accepted widely, airborne laser bathymetry (ALB) typically records the radiometric properties of sensed targets and assists with strip registration, ground (sediment)-type classification, and geometric modeling. Continuous intensity stripping frequently occurs in the ALB-recorded intensity images due to automatic gain control (AGC), an established circuit to compensate for echo power variations caused by distance changes. This situation is exacerbated by the delayed gain control response and inadequate adaptability to target. To address this issue, an adaptive local joint-weighted method for continuous intensity stripping correction is designed. In this method, an index-sharing mechanism is established between intensity, point cloud, emission angle, and return number for rapid localization and intervention of intensity. Unordered intensities are divided into scan line units based on the signal emission period, and subsequent stripped intensity range identification is achieved by assessing the intensity difference between adjacent scan lines. Neighboring scan lines and neighborhood intensities are weighted to jointly reset the stripped intensity values. In the new method, a bidirectional shifting strategy is implemented to attenuate the degradation of the intensity correction accuracy due to the gradual accumulation of correction defects. From the two measurement missions using the ALB, it is concluded that the stripping intensity is mainly distributed in the region of maximum water depth detectable by the sensor and areas with mixed high and low returns. Compared with the original intensity, the mean absolute percentage error (MAPE) and the root mean square error (RMSE) of the corrected intensity decreased by 35% and 40%, respectively, and the variation coefficient of the stripped intensity and the neighboring intensity decreased from 0.4 to 0.18.**

***Index Terms*— Airborne laser bathymetry (ALB), automatic gain control, intensity correction, scanline.**

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I. Introduction

AIRBORNE laser bathymetry (ALB) is a cutting-edge technology in active remote sensing that can be piloted at low cost, high sampling rate, and flexibility over complex regions to perform difficult tasks, such as topographic and geomorphic surveys, coastal environment monitoring, and biological habitat mapping [1], [2], [3], [4]. Compared to traditional acoustic surveying techniques, ALB's unique integration of water and land surveying has earned its recognition as a new dawn in coastal surveying [5]. Beyond the generally accepted spatial location of the target, ALB also provides additional spectral information, a kind of intensity of the backscattered signal, referred to as intensity [6], [7], [8]. ALB intensity linearized to 8-bit/11-bit or other scales [9] has been widely employed in diverse applications, such as strip adjustment, feature extraction, coastal environment monitoring, sediment classification or segmentation, surface modeling, biohabitat mapping, water quality inversion, and natural hazard assessment [10], [11], [12], [13].

Compared to the transmit pulse, the amplitude and pulsewidth of the return pulse are influenced by the wide-ranging array of factors, such as optical–electrical conversion, variations in transmission distance, target reflectivity, and multipath effects, potentially resulting in ranging errors that underlie most intensity unavailable or inaccuracies issues [14]. Weak received signals may result in signal loss due to malfunctioning circuits, while strong signals may cause amplifier circuits to malfunction due to the blockage of the amplifier circuit (nonlinear distortion or overloading issues).

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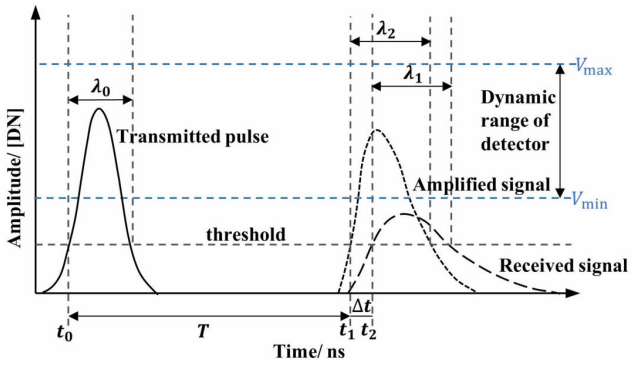


Fig. 1. AGC weakens the ranging error underlying echo variation.

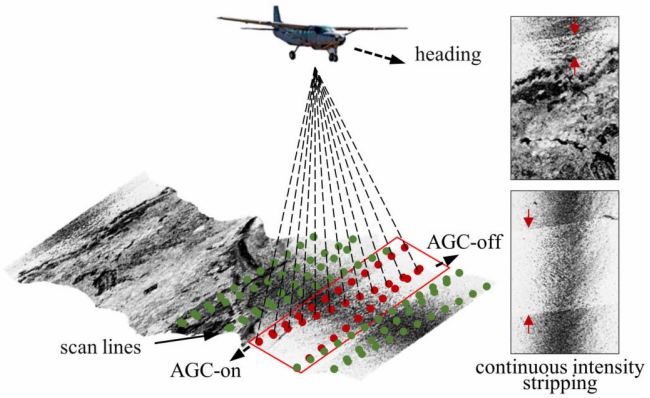


Fig. 2. Continuous intensity stripping in ALB intensity images.

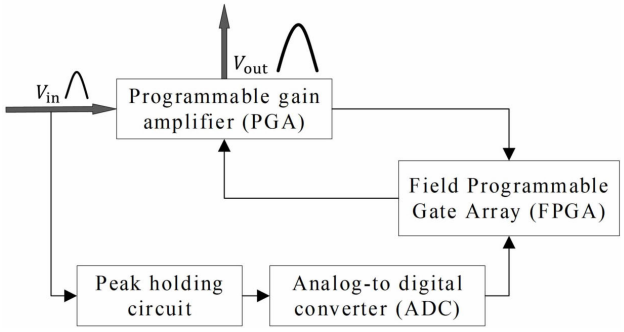


Fig. 3. General model for AGC circuits.

In the early days of radio circuit design, the need to maintain a relatively stable output signal required constant adjustment of the receiver gain to accommodate fluctuations in the received signal amplitude. The need for circuits that maintain a constant output signal level, regardless of variations in the input signal, arose from a given situation [15]. These circuits, initially known as automatic volume control circuits, gradually evolved into automatic gain control (AGC) circuits and became widely accepted [16]. The AGC circuit precisely regulates the echo peak within a predefined range from V_{min} to V_{max}, thus effectively minimizing errors in time jitter and travel [17], [18], [19]. As shown in Fig. 1, the incorporation of the AGC circuit reduces the measurement error caused by the variations in echo pulsewidth and amplitude. Specifically, assuming that the target is detected using the frontal discrimination method when the signal reaches a certain threshold, the circuit will give a logic level as a stop pulse, and the gain effect of the AGC will change the frontal of the pulse, so that the measurement time will be changed from t_2–t_0 to t_1–t_0, resulting in a large deviation in the distance measurement. Moreover, when the target is moving or the device will be used for a certain dynamic range of measurement, the traveling error generated by the moment discriminator used in the echo detection circuit will be different. Such circuits serve as a fundamental component of the new airborne topographic- and bathymetric-LiDAR systems [20], [21]. It amplifies weak echo signals captured at long range while partially reducing signals collected at close range to prevent saturation and distortion [19], [22]. However, Fig. 2 demonstrates that the intensity data is occasionally marred by continuous intensity stripping due to the insufficient gain control response rate and the failure to adapt to targets and backgrounds [23], [24]. These seemingly "scrambled" intensities not only degrade the visual quality of intensity images, but also pose challenges for the application of end-users. In addition, the unstructured, disorganized, and bulky spatial distributions and data volumes exacerbate the difficulty of dealing with stripping intensity. Therefore, if we want to obtain intensity images with natural transitions, it is necessary to eliminate the AGC effect.

A. How AGC Works

AGC is a circuit designed by sensors to control the range of recorded intensities within the radiometric resolution range, specifically between 8 and 150 bits. This range includes floating-point-type intensities ranging from 0 to 1, 8-bit intensity data ranging from 0 to 255 digital number (DN), 10-bit intensity data ranging from 0 to 1023 DN, 12-bit intensity data ranging from 0 to 4095 DN, and 16-bit intensity data ranging from 0 to 65 535 DN [25], [26]. As shown in Fig. 3, the peak keeper, peak detection unit, logic control unit, and programed amplification unit together as the core units constitute the conventional AGC circuit.

Essentially, AGC serves as a negative feedback system that allows the intensity (voltage) of the output signal to be jointly determined by the programmable gain amplifier (PGA) and the gain received by the field programmable gate array (FPGA) chip [27]. Correspondingly, the return pulse is fed into two circuits. In one circuit, a peak keeper detects the pulse peak and holds it for a certain period of time, enabling the peak value, as well as parameters such as carrier frequency, modulation index, or frequency, to be detected by the detector. Immediately afterward, the analog-to-digital converter (ADC) chip quantizes the peak value of the acquired echo pulse and forwards it to the FPGA, which compares it with the "standard voltage" to select the appropriate gain range and relays it to the PGA. In another circuit, the PGA and FPGA chips amplify the pulse signals with different amplitudes according to the received gain information. Ultimately, the AGC circuit controls the output signal within the optimal input range of the constant fraction discriminator (CFD) circuit or other target circuit for signal propagation time calculation [28]. By harnessing this mechanism, the AGC ensures optimal signal detection

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and accurate time measurement for high-performance sensor systems.

B. Overview on the State-of-the-Art of AGC Effect Correction

Despite the recent successful attempts, most existing methods focus on modeling acquisition geometric parameters and environmental factors attenuation compensation, with fewer studies addressing AGC effects [29], [30], [31]. Control system theory has witnessed numerous attempts to characterize AGC systems, ranging from pseudo-linear approximations to multivariable systems [15], [32]. While AGC has been reported for some lidar sensors, the existence of internal processes has rarely been specified, and the impact of AGC has not been fully investigated and modeled [8], [33]. Initially, Korpela [7] sought to demystify the black box of AGC operation by proposing a linear empirical model to capture the impact of AGC and calibrate the correction model's parameters using a reference target. By assessing the influence of AGC compensation on Optech ALTM 3100 and Leica ALS50-II sensors, it was demonstrated that AGC correction can effectively enhance classification accuracy rates by 6%–9% [17]. Subsequently, Korpela et al. emphasized, through monitoring intensity fluctuations, that the AGC effect is nonlinear, and the footprint size varies ride on beam divergence and the scanner–object distance, rendering the linear correction model insufficient [7]. Therefore, Korpela et al. [17] proposed a two-parameter (range and AGC) calibration model for the ALS50 sensor using calibration equations to minimize intensity variations, as follows:

$$I_{\text{cal}} = (R/R_{\text{ref}})^a \cdot I_{\text{obs}} + (b \cdot I_{\text{obs}}) \cdot (127.5 - \text{AGC}) \quad (1)$$

where I_{cal} refers to the calibrated intensity value, I_{obs} represents the measured intensity value, R denotes the propagation distance, R_{ref} signifies the average reference range, AGC indicates the recorded gain value (GV), and a and b are the calibration parameters.

Vain and Kaasalainen [21] and Vain et al. [26] analyzed the intensity data recorded in two states of the AGC function: AGC-on and AGC-off. Based on their findings, they posited that the impact of AGC is modulated by two factors: a scaling factor and an offset factor. Drawing from this observation, they proposed a segmented linear function to describe the conversion equation between the recorded intensity value I_{on} in the AGC-on mode and the intensity value I_{off} recorded in the AGC-off mode

$$I_{\text{off}} = a + b \cdot I_{\text{on}} + c \cdot I_{\text{on}} \cdot \text{AGC}. \quad (2)$$

To develop the aforementioned model, two key components are required: the GV of each intensity sample and the pretrained model parameters. However, it should be noted that obtaining the above prior information is often a challenging task in practice, particularly since GVs are often inaccessible for most commercial laser scanners. Moreover, the model parameters exhibit a strong correlation with the sensor model, operating parameters, and feature class, preventing their adaptation and utilization across different tasks. In theory, (2) defines parameter a as having specific values for different target types. Specifically, it equals 2 for uniform targets that exceed the footprint, 3 for linear objects like wires, and 4 for individual large scatterers [33], [34]. Determining these parameters often necessitates a customized experimental setup and an optimization search.

Rohde et al. have previously emphasized that the AGC effect perturbs intensity by ±27% [16], making it unignorable in intensity mapping. GV is constantly fluctuating under normal operating conditions, which inevitably results in intensity changes, even if the targets are the same (e.g., asphalt, bare rocks, and roads). Although numerous AGC correction approaches exist for airborne topographic laser scanners (ATLSs), limited attention has been paid to ALB. Compared to ATLS, AGC during ALB operation is characterized by increased "activity" and "complexity," due to its primarily coastal working environment that comprises a high density of both low-reflective targets (such as water surfaces) and high-reflective targets (such as reefs and sandy beaches). The sluggish response and inadequate adaptability to targets and backgrounds render the AGC effect on ALB intensity images particularly salient. Although the AGC's fundamental principle is straightforward, nonlinear equations that arise during analysis often defy resolution in most cases [36]. Furthermore, existing AGC correction models' parameters are nonuniversal and challenging to implement. To address the above issues, this article proposes an adaptive local joint-weighted intensity correction model that enables point-by-point intensity correction without reliance on any prior information or parameters. The main contributions of this article are as follows.

1) The root causes of the AGC effect and the characteristics and distribution patterns of the continuous intensity stripping problem are revealed.
2) An adaptive local joint-weighted intensity correction approach is proposed to address the continuous intensity stripping problem triggered by the AGC effect.
3) The intensity data are intelligently segmented in scanline units and a uniform sampling-based Kolmogorov–Smirnov (K–S) test intensity stripping scanline identification approach is designed. This approach saves arithmetic space compared to the point-by-point-based one.
4) Comparative experiments are conducted to verify the rationality and advantages of the new model's bidirectional shift strategy and the performance of local joint weighting for intensity correction.

In summary, this article presents an innovative method to address the challenges of AGC correction in ALB data. The proposed adaptive local joint-weighted intensity correction model offers a promising approach to enhance the accuracy of ALB intensity mapping and expand its practical applications.

II. Data

The data for this study were gathered from two islands in China using the Aquarius system developed by Optech. This compact and innovative system offers a state-of-the-art solution for mapping both coastal and inland waterways. It boasts a dynamic measurement range that spans from 0 to 4095 DN,

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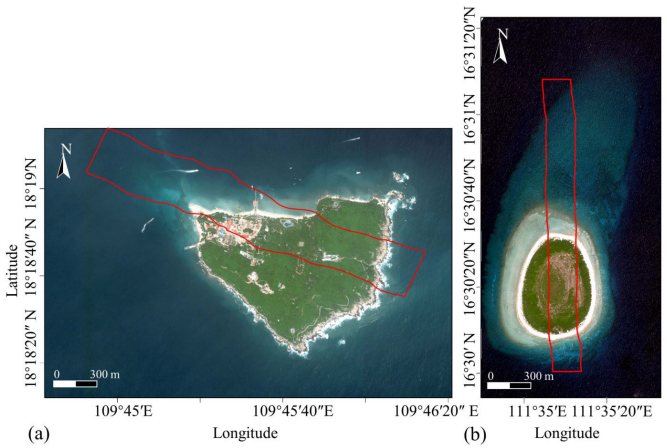


Fig. 4. Location of ALB data collection: (a) Wuzhizhou Island and (b) Zhiyuan Island.

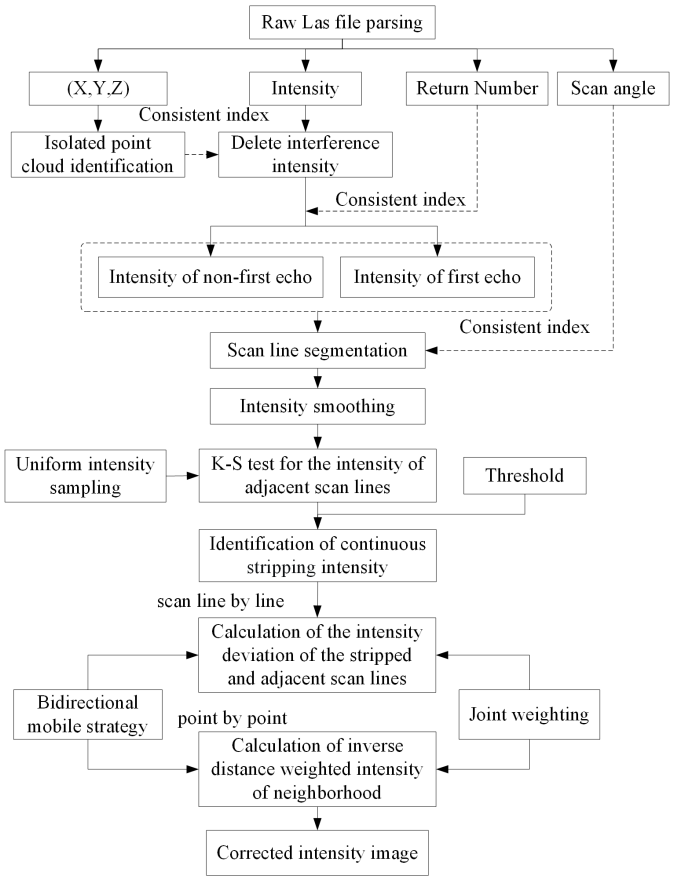


Fig. 5. Workflow diagram of adaptive local joint weighting for continuous intensity stripping correction.

with an impressive 12-bit resolution. Surveying line 1 was situated on Wuzhizhou Island, collected in 2012 [Fig. 4(a)], and spans approximately 2300 m in length, with a flight height of 300 m and a scanning width of 25°, resulting in approximately 2.05 million sample points. The red closed curve indicates the coverage range of the strip. Surveying line 2 was collected on Zhiyuan Island in 2013 [Fig. 4(b)] and measures approximately 2200 m in length, with a flight height of 300 m and a scanning width of 22°, yielding over 1.52 million sample points collected.

III. Methods

A. Methodological Workflow

The adaptive local joint-weighted ALB continuous stripped intensity correction process comprises three steps, as illustrated in Fig. 5: preprocessing of intensity data, identification of continuous stripping intensity, and local joint-weighted intensity correction. This approach not only enhances data accuracy, but also fosters a deeper comprehension of the target area. Improved intensity data processing and correction techniques yield more dependable information to underpin subsequent target analysis and decision-making.

1) Intensity Data Preprocessing: Intensity data preprocessing involves two core steps: removing interference and segmenting intensity data. Parameters such as point cloud coordinates, intensity, return number, and emission angle are established with index numbers. Coordinates help remove noise, and intensity is divided into first-echo and nonfirst-echo intensity based on the return number parameter.

2) Continuous Stripping Intensity Region Identification: Preprocessed intensities undergo segmentation into scanline units based on emission angle and filtered to present intensity noise interference. Specifically, the K–S test is performed on neighboring scanline intensities to identify stripped regions that may exist during intensity data acquisition for subsequent correction processing.

3) Stripping Intensity Correction: Point-by-point intensity correction is achieved using local joint weighting and a bidirectional shift strategy. Local joint weighting optimizes the tradeoff between preserving stripped intensity details and minimizing intensity deviations from neighboring scanlines.

B. Intensity Data Preprocessing

To assess stripping intensity, it is imperative to establish appropriate measurement conditions before any corrections can be implemented. The producer's requirement for intensity lies in its capacity to convey the target's attributes. In most cases, the initial echo's potency is sufficient. However, for underwater scenarios, the seafloor echo intensity indicates the seabed sediment types. Hence, considering the return number in intensity correction is crucial. Intensity data preprocessing mainly involves interference intensity rejection and intensity segmentation. Ji et al. [20] confirmed that instrument noise generates invalid intensities, corresponding to a generally low-intensity range from 30 to 50 DN and high altitude (more than 120 m). The threshold is set to any value that falls within the required range, being less than the minimum height of the isolated points cluster and greater than the maximum height of the normal sampling point. Invalid intensities can be identified and excluded by setting an elevation threshold.

Typically, a pulse emission will have a corresponding emission parameter (angle of incidence), target location (coordinates), attribute information (intensity), as well as waveform and potential information derived from the waveform, such as the number of echoes. This information is theoretically

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one-to-one, and when one parameter is considered as target data, all other parameters can be used as auxiliary data. To achieve the objective, the article proposes an index-sharing mechanism that assigns index numbers to corresponding intensities, coordinates, emission angles, and return numbers, according to a one-to-one correspondence. This approach facilitates efficient linkage of individual parameters. To detect and suppress interference intensity, an elevation threshold is set, specifically by filtering the index numbers of the sampling points that exceed the threshold and searching and culling the intensity data that have consistent indices. The remaining intensity is then categorized into first-echo and nonfirst-echo based on the return number.

C. Identification of Continuous Stripping Intensity

Intensity, similar to the point cloud, is characterized by disorder, making it challenging to localize and correct stripped intensity in cluttered data. Determining the portion of data belonging to the stripped intensity is the initial challenge that needs to be addressed. Visually, stripped intensity exhibits a significant offset from the correct intensity, assumed to result in intensity sets belonging to different distributions. Subsequently, the separation of these intensity sets can be achieved through a comparison of their spatial distributions or statistical patterns. The Kolmogorov–Smirnov (K–S) test, a widely utilized statistical analysis method, compares an observed frequency distribution with a "standard" frequency distribution to assess the significance of the observed distribution pattern [27]. In this analysis, the data to be statistically analyzed (the scan line intensities to be evaluated) are compared to another set of "standard" intensities (the closest scan lines that have been evaluated) to determine the deviation between the two. This deviation is then employed to quantify the extent of difference between the two datasets.

In the standard K–S test, it is assumed that the frequency distributions of the two datasets being compared are similar. If the difference between the two frequency distributions is small, there is a likelihood of such a difference occurring by chance. However, if the difference is significant, the likelihood of chance occurrence is reduced. The traditional method of point-by-point intensity difference judgment is impractical due to the high computational requirements. Given the time-continuity characteristics of stripping intensity occurrences, we propose a simplified approach to enhance data processing efficiency. Specifically, we set the unit of intensity difference judgment as a scanning line. By capitalizing on the emission angle's periodic variation, we have segmented the emission angle parameters. By utilizing the shared index number, the intensity data are divided into small datasets corresponding to complete scan lines. To compress computational effort and mitigate noise interference, it is essential to filter and resample intensity data before calculating intensity deviation for neighboring scan lines. Gaussian filtering, a linear smoothing technique that uses a Gaussian kernel for weighted averaging, ensures good noise removal and is effective for both Gaussian and pretzel noise and is therefore employed on scanline-by-scanline filtering.

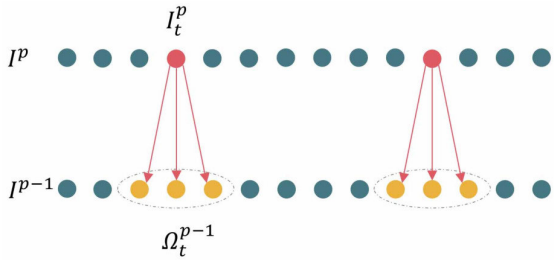


Fig. 6. Neighboring scan line intensity extraction based on uniform sampling.

In the direction of flight, the intensity discrepancy between the pth scan line and the $p-1$th scan line is assessed. Uniformly sampled intensity data $I^p = (i_1^p, i_2^p, i_3^p, \ldots, i_n^p)$ are selected from the pth scan line, and the neighborhood intensity set Ω searched by nearest distance is averaged for each intensity data to obtain the corresponding intensity data for the $p-1$th scan line $I^{p-1} = (i_1^{p-1}, i_2^{p-1}, i_3^{p-1}, \ldots, i_n^{p-1})$, as shown in Fig. 6. Empirical cumulative probability functions $F_p(i)$ and $F_{p-1}(i)$ are then calculated for I^p and I^{p-1}

$$F_p(i) = \frac{1}{t}\sum_{j=1}^{t} \mathrm{card}\left(i, i_j^p\right) \tag{3}$$

$$F_{p-1}(i) = \frac{1}{t}\sum_{j=1}^{t} \mathrm{card}\left(i, i_j^{p-1}\right) \tag{4}$$

where card$(\cdot)$ represents a one-dimensional indicator function that returns a value of 1 if the input is true, and 0 otherwise. The maximum distance $D_{p,p-1}$ between the empirical cumulative distribution functions $F_p(i)$ and $F_{p-1}(i)$ serves as a criterion to determine the significance of intensity differences between consecutive scan lines

$$D_{p,p-1} = \max |F_p(i) - F_{p-1}(i)|. \tag{5}$$

The pth scan line is considered to be the stripped intensity when D is greater than or equal to the threshold value ε. By iterating over all scan lines, the stripped intensity continuum can be identified.

D. Adaptive Local Joint Weighting

To apply the intensity product, the stripped intensity must be corrected. Replacing the original intensity with interpolated intensity may result in a loss of detailed information. To address the challenges of stripping intensity, a correction approach is proposed that considers the detail of the stripped intensity data as well as the intensities of neighboring scan lines, as illustrated in Fig. 7. The nonlinear nature of the stripped intensity the modification methods reliant on fixed strength offsets incapable of delivering precise correction results. In the novel approach, the target intensity data is jointly determined by the neighboring scanline intensity and the neighborhood intensity. Specifically, the corrected intensity $I_j^{p'}$ corresponding to the jth intensity of the pth stripping intensity scanline is obtained by jointly weighing two parts: one part is the mean value of the neighboring intensity Γ_j^p of i_j^p, and the other part is the combination of the intensity

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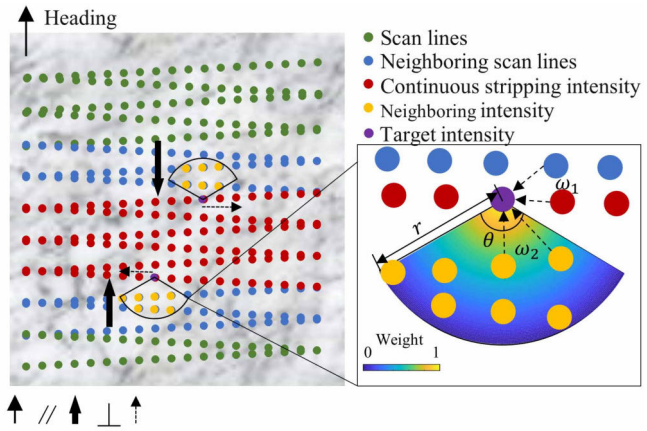


Fig. 7. Schematic of intensity correction with adaptive local joint weighting.

deviation statistic Δ^p of i_j^p and its located scanline and the neighboring scanline

$$I_j^{p'} = w_1\left(i_j^p + \Delta^p\right) + w_2\Gamma_j^p \tag{6}$$

$$\Delta^p = \bar{I}^{p-1,p-s} - \bar{I}^p \tag{7}$$

$$\Gamma_j^p = \sum_{t=1}^{k} \omega_t \Omega_j^p \tag{8}$$

$$\omega_t = \sum_{t=1}^{k} \left(\frac{1}{d_t} \left(\sum_{t=1}^{k} \frac{1}{d_t} \right)^{-1} \right) \tag{9}$$

where $I^p = (i_1^p, i_2^p, i_3^p, \ldots, i_n^p)$ is the set of intensity data for the pth scan line, n is the number of intensity values, $\bar{I}^p$ is the mean value of I^p, and $\bar{I}^{p-1,p-s}$ is the mean value of the set of intensity data, $\bar{I}^{p-1,p-s}$, for the neighboring s complete scan lines from $p-1$ to $p-s$. Γ_j^p is the inverse distance-weighted average of the intensities $\Omega_j^p = (i_1, i_2, \ldots, i_k)$ in a sector with i_j^p as the origin and θ as the angle of center of the circle r as the radius. Here, k is the number of intensity values, and the center of the circle is oriented parallel to the heading. The weight ω_t is inversely proportional to the distance d_t of the interest point $\Omega(i_t)$ from the center of the circle i_j^p, and ω_t sums to 1. Similarly, w_1 and w_2 are for the weights, with the sum equal to 1. To determine the assignment of the weights, we constructed the objective function g, which is defined as follows:

$$g = \min \left\| \sum_{t=1}^{M} \left| I_j^{p'} - i_j^{p'} \right| \right\| \tag{10}$$

where $|I_j^{p'} - i_j^{p'}|$ is the absolute value of the deviation of the corrected intensity $I_j^{p'}$ from the mean value of the neighboring intensities $i_j^{p'}$. The determination of w_1 and w_2 also needs to be done by uniform sampling. The intensity in scan line p is downsampled to obtain M sampled intensities $i_t^p, t = 1, 2, 3, \ldots, M$. Finally, the optimal w_1 and w_2 values are determined using the least-squares method.

E. Bidirectional Movement Strategy

To solve the issue of decreasing correction effectiveness due to insufficient intensity correction and gradual transfer accumulation, this article proposes a bidirectional moving correction strategy. This strategy involves initiating the intensity correction process simultaneously from both ends of the striped intensity and gradually moving toward the center, ultimately achieving complete correction of the intensity data, as illustrated in Fig. 7. By sequentially applying this operation to all stripped scan lines and intensities, we obtain the corrected intensity image.

IV. Results

A. First- and Nonfirst-Echo Intensity Data

Based on the self-developed Las data analysis and processing software SplitLAS V1.0, the parsing of the Las file is achieved to obtain each parameter, including coordinates, intensity, incidence angle, and return number. An index-sharing mechanism is established to enable the integrated operation of each parameter, including identification, rejection, and correction. To eliminate the influence of invalid intensity data on the assessment of stripping intensity, a height threshold of 120 m is set to identify and reject this type of intensity. Other parameters, such as corresponding coordinates and incidence angles, are also rejected. When laser light passes through a target, such as water or vegetation, a significant loss of energy occurs due to refraction and scattering. Therefore, the magnitudes of different echo intensities vary and cannot be corrected equally. To obtain consistent intensity data, we further classify the first return and nonfirst return intensities by return number. If the return number parameter is invalid or missing, we can achieve intensity segmentation by analyzing the full waveform data.

In Fig. 8, raw intensity data for two study areas are presented in Fig. 8(a) and (d), along with the first-echo intensities in Fig. 8(b) and (e) and nonfirst-echo intensities in Fig. 8(c) and (f). The results indicate that invalid intensity distributions are mainly concentrated in the central region and removing them enhances the intensity image quality. The stripped intensity distribution is consistent across first-echo intensity and nonfirst-echo intensities, suggesting that laser emission disturbed by the AGC effect improperly adjusts the intensity, affecting all echo intensities. A comparison of Fig. 8(e) and (f) confirms the need to treat first/nonfirst-echo intensities differently as stripped intensity data bias varies in different echo layers.

B. Continuous Stripping Intensity Recognition

The intensity data are segmented and numbered into scan lines based on the periodic variation of the emission angle. To mitigate the impact of noise, Gaussian filtering is applied to each scan line, effectively rejecting isolated intensity values. A fixed number of intensity samples (30, approximately 9% of the total number of intensities for a single scanline) are taken for each scan line. For each scan line, a fixed number of intensity samples are taken. By setting $\mathcal{E}$ to 0.2 and performing a K–S test on the sampled intensities, the boundaries of continuous stripping intensities are determined, defining the range of stripping intensities. The stripping intensities of first-time and nonfirst-time-echo intensity data are then identified separately.

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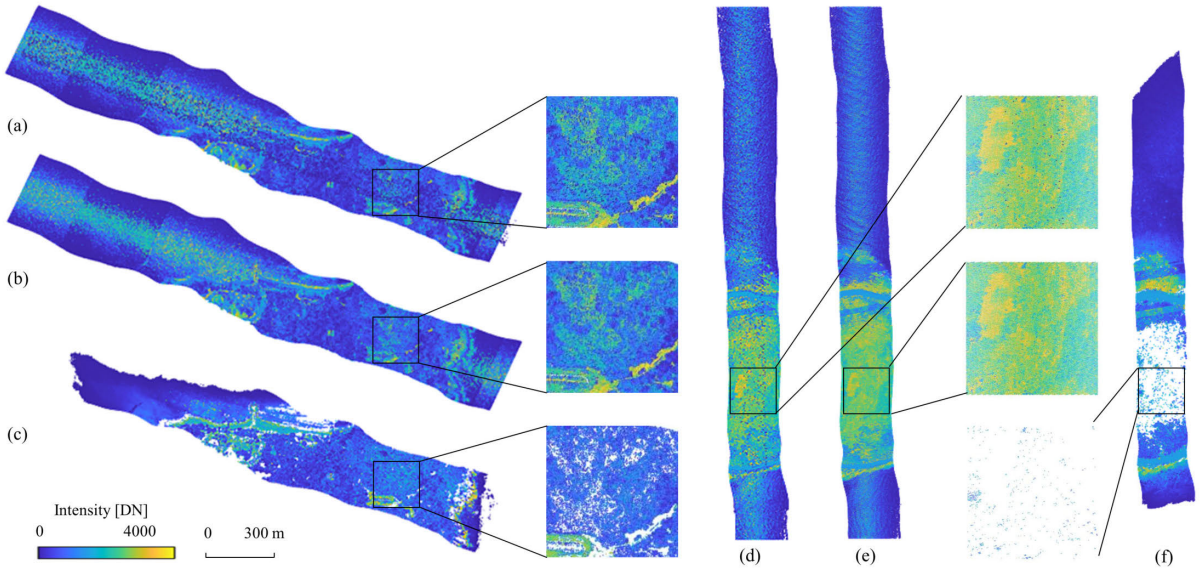


Fig. 8. Intensity images obtained for different return (first and nonfirst echo) signals. (a)–(c) Correspond respectively to the raw, first-echo, and non-first-echo intensity maps of region A. (d)–(f) Correspond respectively to the raw, first-echo, and non-first-echo intensity maps of region B.

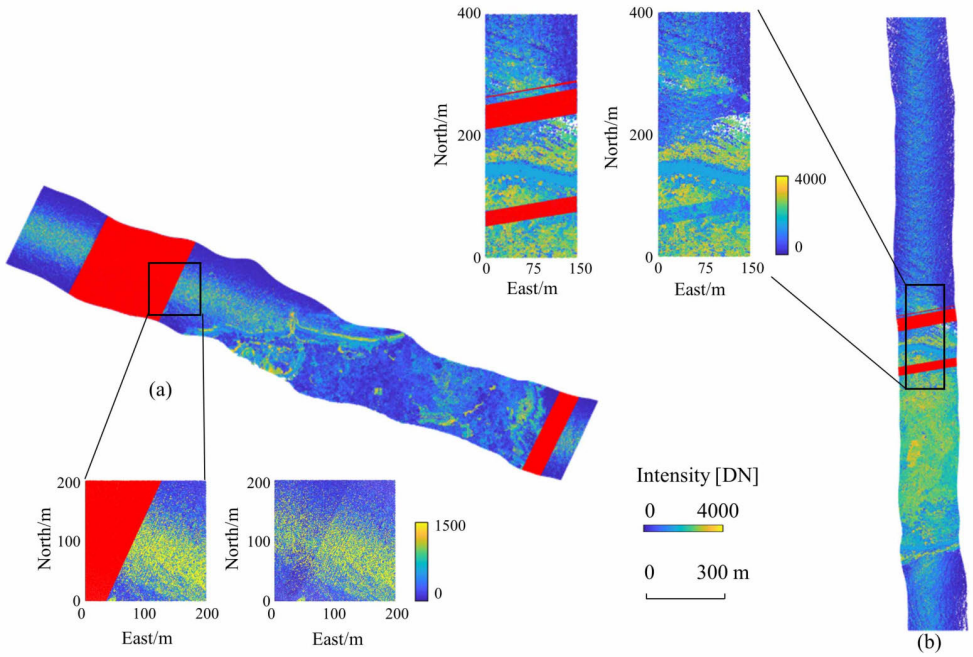


Fig. 9. Continuous intensity stripping scan line identification results: (a) region A and (b) region B.

In the process of identifying stripping intensities, we find that as the water depth increases, the intensity of seabed returns weakens, leading to a smaller deviation of stripping intensity from neighboring intensities and increased difficulty in identification. In contrast, sea surface return intensities have better stripping intensity identification. However, in shallow water areas, the mixing of the water column and bare rocks makes it difficult to determine the range of stripping intensities for first-return intensity data, while nonfirst intensity data show excellent identification accuracy. Therefore, we combine the stripping intensity identification results from different return numbers to obtain more accurate identification results.

Through analysis, two stripping intensity areas are identified in study area A, totaling 256 intensity anomaly scan lines. These areas are comprised of 221 and 35 scan lines, respectively. Similarly, in study area B, three stripping intensity zones are identified, totaling 41 scan lines. These zones have 2, 24, and 15 scan lines, respectively. The distribution of stripping intensity is shown in Fig. 9, represented by the red point cloud. Examination of the two zoomed-in areas reveals a distinct difference in stripped intensity compared to neighboring intensities. The distribution range of the identified stripping intensity scan lines appears visually consistent with the intensity data exhibiting intensity deviations. This further

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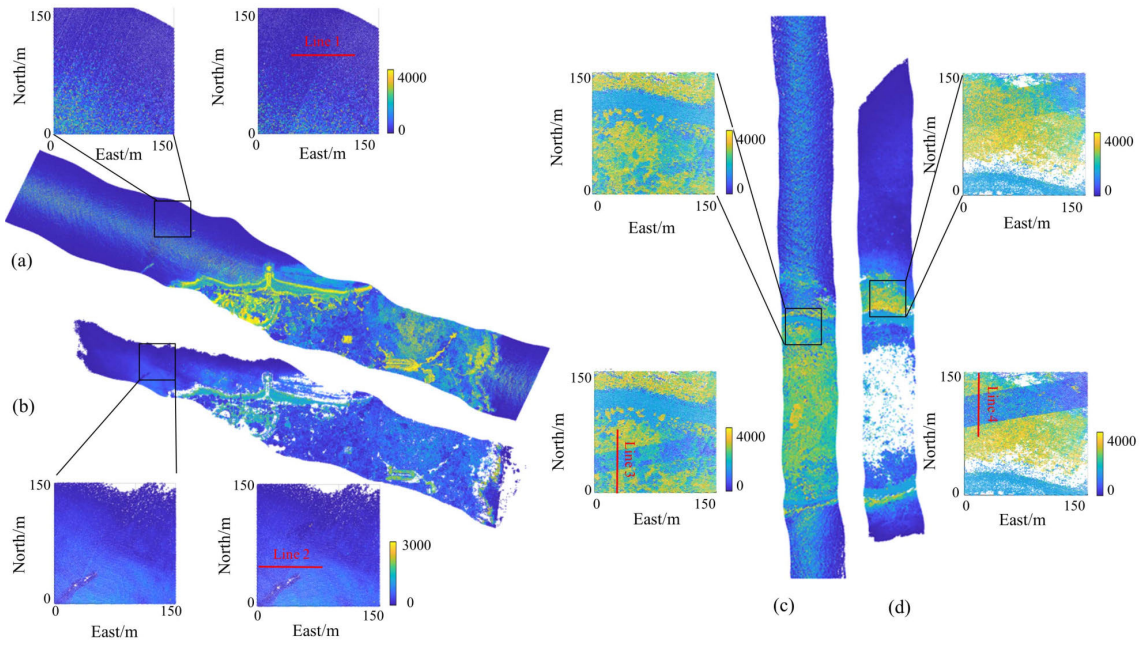


Fig. 10. Intensity image after correction of continuous stripping intensity. (a) Area A—corrected first-echo intensity results. (b) Area A—corrected nonfirst-echo intensity results. (c) Area B—corrected first-echo intensity results. (d) Area B—corrected nonfirst-echo intensity results. The corresponding zoomed-in plot displays the uncorrected intensity data.

establishes the precision and reliability of our stripped intensity identification approach.

C. Evaluation of Intensity Correction Accuracy

The parameter s for the scan line is set to 4, the number of uniform samples t is set to 20, and the sector center angle θ is chosen to be 120° with a radius size of 2.5 times the spatial resolution of the intensity data. Neighboring scan lines are controlled within a range of approximately 2 m, serving as the basis for the determination of s, as the binding force diminishes with increasing distance. We employ a local joint-weighted correction method based on bidirectional shifts for intensity correction in each anomaly region. This approach enables control over the neighboring five scan lines, which are subsequently resolved. The final intensity correction results with different numbers of returns are obtained. Fig. 10(a) and (b) presents the first-echo and nonfirst-echo intensity correction results for region A, respectively, while Fig. 10(c) and (d) shows the corresponding results for region B. We select a rectangular area of 150 × 150 m from each of these four intensity images to show the details before and after intensity correction. The square-boxed areas are zoomed in to reveal the results after stripping intensity correction (leaning to the left and up), with their counterparts (leaning to the right and down) representing the preprocessed intensity data. Visually, the corrected intensity image displays a more natural transition of feature intensities and clearer boundaries of feature classes.

An area of 80 × 2 m is selected in each of the four local zoomed-in regions and designated as Line 1–Line 4. Fig. 11 illustrates the profile results before and after intensity correction for Line 1–Line 4, with red indicating intensity before correction and blue indicating intensity after correction.

TABLE I
COMPARATIVE ANALYSIS ON THE EFFECT OF INTENSITY CORRECTION

Area		Line 1	Line 2	Line 3	Line 4
Before correction	MD(DN)	445	456	3629	3459
	MAPE	0.019	0.037	0.309	0.301
	MD(DN)	75.913	105.372	855.435	828.852
	CV	0.576	0.203	0.401	0.423
Scale factor -based method	MD(DN)	331	323	2320	2409
	MAPE	0.017	0.029	0.226	0.179
	MD(DN)	52.348	79.199	673.310	533.207
	CV	0.378	0.193	0.306	0.199
Proposed method	MD(DN)	220	301	2384	1916
	MAPE	0.016	0.021	0.195	0.169
	MD(DN)	44.780	62.109	560.397	475.782
	CV	0.215	0.111	0.223	0.171

Generally, the stripping intensity is slightly lower compared to the neighboring normal regions. However, the intensity deviation varies across different regions, with the intensity deviation of Line 2 being smaller than that of Line 3, which is smaller than that of Line 1, which is smaller than that of Line 4. To quantitatively evaluate the effect of intensity correction, the corrected intensities obtained based on the method of this article were analyzed in comparison with the original intensities and the correction method based on the scaling factor. The scaling factor-based approach seeks to align the maximum and minimum intensities of neighboring scan lines, scaling the intensities uniformly within the interval. Samples from the 20-m anomaly zone and the neighboring 20-m normal intensity were selected for accuracy analysis. The intensity in the normal zone is considered as the true value, and maximum deviation (MD), mean absolute percentage error (MAPE), root-mean-square error (RMSE), and coefficient of variation (CV) are calculated.

Table I presents the intensity statistics before and after correction. A comparison of the intensity data reveals that

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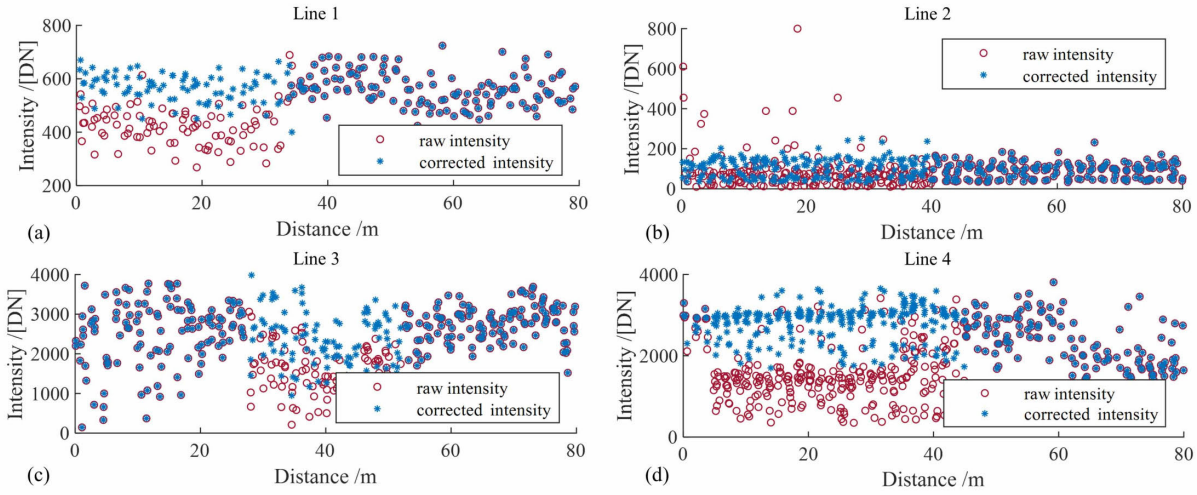


Fig. 11. Comparison of Line 1–4 intensity before and after correction. (a)–(d) Intensity of Line 1–4 before and after correction, respectively.

the continuous stripping intensity correction method based on adaptive local joint weighting effectively eliminates the AGC effect. The four parameters of MD, MAPE, RMSE, and CV are significantly reduced in the corrected intensity images. The mean value of the CV decreases from 0.40 to 0.18. Comparison with the scale factor-based intensity correction method confirms that linear scaling fails to eliminate the nonlinear effect of stripping intensity deviation. The corrected intensity data retain detailed features while maintaining a natural transition with neighboring intensities. A comparison experiment verifies the effectiveness of the proposed method, providing data assurance and technical support for subsequent intensity data feature mining and applications.

V. Discussion

A. Distribution Pattern of Stripping Intensity

We observe that the stripped intensity data are primarily distributed in two scenarios. In the first scenario, close to the maximum detectable water depth, the laser scanner experiences weak return energy, leading to activation of the GV. After crossing this region, only the surface return signal is received, leading to a significant deviation in the intensity data close to the bathymetry detection limit compared to neighboring areas. In the second scenario, areas with high return targets (exposed reefs and towering trees) or low return targets (waters) are encountered, where the GV of the AGC changes frequently and complexly. When the laser scanner is over a water object, the GV rises. When the laser beam strikes a ground object again (suddenly), the GV remains continuously elevated or reaches a maximum value. Due to the dependence of intensity value on GV, the intensity varies before and after encountering the water object. Similarly, when the laser falls on tall trees, the return energy is higher, the GV increases inversely, and the intensity is suppressed. Regions with frequent occurrences of high/low return targets tend to produce striped intensities. In region a, striped intensity regions are all distributed in the critical region of water depth detection, while in region b, striped intensity regions are distributed in the nearshore shallow water region with reefs and in the wooded region.

B. Characterization of Stripping Intensity

To estimate intensity changes due to AGC effects, three stripped intensity datasets in the study area are randomly selected and compared with neighboring normal intensity. Fig. 12 profiles the intensity of these three regions along the scan line direction, where blue represents normal intensity and red corresponds to stripped intensity scan lines. The trends of the two scan lines are similar, but there are clear intensity differences. This approximate change trend indicates that the intensity data under interference by AGC remains the ability to express the target attributes.

Specifically, Fig. 12(a) demonstrates that the intensity differences corresponding to different intensities of the same scan line do not exhibit an obvious linear relationship with intensity. Fig. 12(b) reveals that even for the same intensity of the same scan line, which is equal to the randomly chosen intensity value, indicated by the black dashed line, there are obvious differences in the corresponding intensity deviations, with both positive and negative differences. For Fig. 12(a)–(c), the intensity values equal to 1000 are selected and compared, intersected at the green dashed line, revealing that the intensity deviation corresponding to the same intensity of different measurement lines varies. In summary, AGC interference preserves intensity details, however, significant over or undercompensation differences are nonstandardized and independent of angle/intensity. Therefore, a single intensity deviation or specific scaling factor cannot meet accuracy requirements.

C. Bidirectional Movement Strategy Effects

In theory, the adaptive local joint-weighted continuous stripping intensity correction proposed in this article relies on the intensity of neighboring lines or neighborhoods defined as "accurate intensity." However, if the correction under this logic is insufficient or excessive, the resulting bias will accumulate

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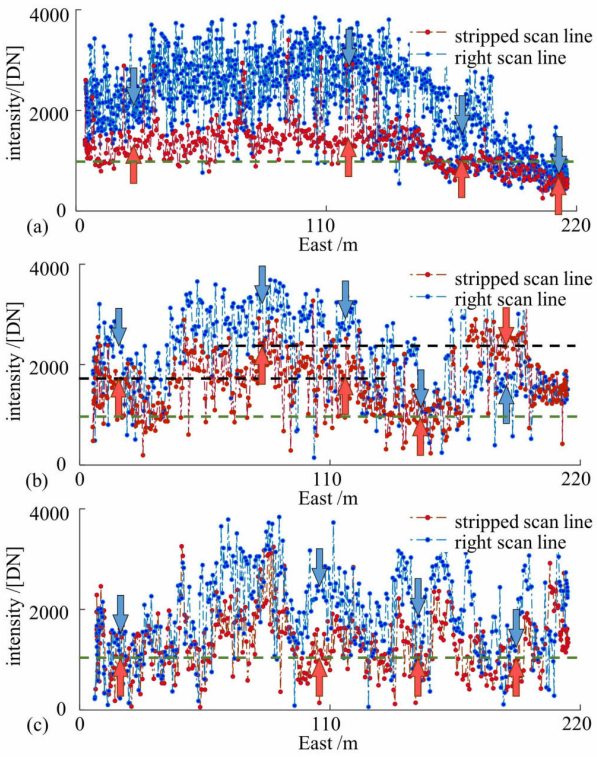


Fig. 12. Intensity of the stripped intensity scan line versus the neighboring normal scan line. (a)–(c) Intensity of three regions along the scan lines, where blue represents the normal intensity and red corresponds to the stripped intensity.

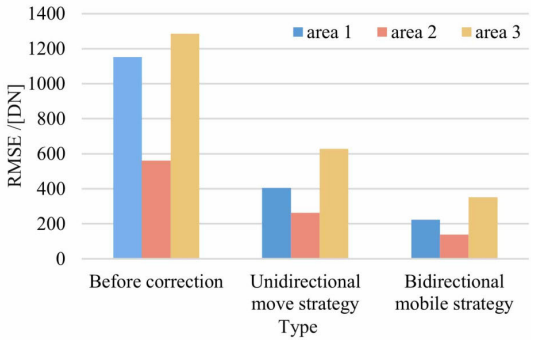


Fig. 13. Analysis of the effect of the bidirectional moving strategy.

as the intensity correction progresses from line to line, resulting in a decreasing effect of intensity correction. To solve this problem, we design and incorporate a bidirectional moving strategy. To evaluate its effectiveness, we select three intensity-stripping boundary regions and compare the intensity correction results under the bidirectional moving strategy with the original intensity data and the intensity data obtained by the unidirectional moving strategy. The comparison results are shown in Fig. 13.

In the analysis, three pairs of scan lines are selected for comparison, and 15 pairs of neighboring intensities are uniformly sampled. The intensity of neighboring scan lines in the normal intensity region is considered the true value. The RMSE of the intensity data obtained by different methods is used as the evaluation parameter for the correction effect. In Fig. 13, we can see that the intensity correction results under the bidirectional moving strategy are more accurate (RMSE are less than 351 [DN]) and closer to the original intensity data. The bidirectional moving strategy's effectiveness in constraining error accumulation is demonstrated by its superiority over the unidirectional moving strategy and its original strength. Overall, the bidirectional moving strategy is designed to address the issue of accumulated bias and to improve the accuracy of intensity correction. It significantly outperforms the unidirectional approach as it effectively addresses the problem of accumulated bias and results in more accurate and reliable intensity correction results.

VI. Conclusion

Reliable optical communication systems commonly employ AGCs for laser amplifiers to mitigate the impact of signal polarization drift, input power fluctuations, temperature variations, and device aging. Nevertheless, the wide amplitude variations in the output caused by AGC circuitry can interfere with the signal and result in information loss or unacceptable system performance. These unfavorable effects are particularly prominent in the ALB. In our investigation of the intensity stripping caused by AGC effects in ALB, we draw the following conclusions.

· AGC implementation results in an internal process running according to a predetermined program that changes the recorded intensity value of the target. This change makes the intensity compensation values difficult to track, present nonnormalized, and independent of the sensor geometry parameters. Furthermore, it was discovered that despite the stripped intensity data produced by the AGC effect being significantly distinct from neighboring regions' intensity data, the AGC effect preserves the intricate details. This bias is complex, with different intensity biases in different regions and even in the same region with different return numbers. Therefore, the AGC effect diminishes the reliability of intensity data for applications such as ground object classification and segmentation, necessitating its correction.

· The AGC algorithm cannot anticipate features and boundaries, resulting in delayed GV adjustments and abrupt intensity variations. To address these issues, this article proposes a local joint-weighted intensity correction method that considers preprocessing of intensity data, identification, and correction of intensity peeling areas. Compared to point-by-point/block-by-block intensity judgment, the scanline-based peeling intensity identification mode simplifies the computation process and accurately obtains peeling intensity boundaries. Local joint-weighted intensity correction preserves details of the original intensity record while correcting intensity deviations from neighboring intensities. Furthermore, the bidirectional shift strategy effectively mitigates the accumulation of intensity correction errors compared with the traditional unidirectional shift strategy. The proposed method effectively compensates for intensity "mutations" caused by AGC anomalies, controlling the variation coefficient of corrected neighboring intensities within 0.23. Compared to precorrection results, MAPE is reduced by 35% and RMSE is reduced by 40.

· Essentially, LIDAR's core is laser ranging, leading to circuit design preferences for precise point computation. Future AGC design concepts should consider the interference of AGC

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effects on intensity data. Additionally, the AGC loop's multiposition switches result in complex circuits and high power consumption, rendering them unsuitable for multichannel use. This is a problem that needs to be addressed in the future.

Sensor manufacturers have not disclosed the working mechanism of AGC, leading to insufficient research and modeling of its effects. The technology to eliminate AGC interference remains problematic. Therefore, future research should focus on revealing the "black box" of AGC operation, breaking technical barriers, fundamentally eliminating its interference, refining the ALB intensity correction model, and obtaining precise target attribute parameters.

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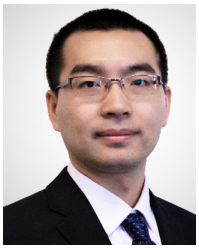
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